

vulnerabilities by focusing on network nodes whose failure could severely disrupt service. By integrating the network model with quantum computing for analysis, the initiative seeks to boost network reliability and service quality. Kipu Quantum's quantum algorithm enables thorough inspection of R's network using quantum annealers and neutral-atom quantum computers for effective data analysis. The approach utilised D-Wave's quantum annealers for preliminary analysis and QuEra's quantum computer for a detailed study, with AWS Braket and D-Wave providing cloud platform support. This application of quantum computing in network management promises significant improvements in fibre network operations, with quantum technology poised to address more complex infrastructures by 2025.

Algorithms developed for real-time learning in autonomous devices

Researchers in the US Department of Energy's Argonne National Laboratory have developed algorithms for continuous learning. They point out the quest for real-time adaptation necessitates advanced algorithms that, in turn, require hardware accelerators to be more powerful in terms of energy, memory, and flexibility. These accelerators need to be compact, energy-efficient, and able to adjust their resource use over time to learn new information without losing previously acquired knowledge, a challenge known as "catastrophic forgetting." The team outlines essential attributes for AI accelerators to facilitate lifelong learning: the ability for on-device learning, flexible resource management, model recoverability to ensure consistent performance amid ongoing learning, and efficient integration of new and existing knowledge. Meeting these demands is a significant hurdle, marking the beginning of efforts towards creating autonomous devices with the capacity for uninterrupted, real-time learning.

Management of rooftop solar panels made easy by new software

Researchers at SLAC National Accelerator Laboratory have developed Solar Data Tools to improve the management of rooftop solar panels. The software, which analyses data from 1000 solar installations, monitors power output in five-minute intervals over years to reveal how weather, shading, and damage impact power generation. It uses signal decomposition to filter out noise and focus on performance metrics, helping owners spot damage or inefficiencies. Initially designed for technically skilled users, it creates graphs to swiftly track and address system issues. Efforts are underway to broaden its accessibility through a user-friendly mobile app and a simple interface, making solar data analysis understandable for non-experts.

A polymer electrolyte membrane extends fuel cell lifespan

Researchers from Incheon National University and Harvard University have innovated a polymer electrolyte membrane



Scientists have successfully engineered a fatigue-resistant electrolyte membrane that exhibits resistance to cracking and significantly extends the lifespan of fuel cells (Source: oakridgelabnews from Openverse <https://openverse.org>)

for fuel cells that exceptionally resists fatigue, addressing a critical need for durability in real-world conditions. By employing an interpenetrating network strategy, this membrane evenly spreads stress, combining Nafion—a leading proton-conducting electrolyte—with

perfluoropolyether (PFPE) to form a resilient blend. This composition slightly lowers electrochemical efficiency but significantly boosts wear resistance and longevity. Testing showed that a membrane with a 50% PFPE blend not only maintains satisfactory electrochemical performance but also enhances fatigue resistance by 175% and extends fuel cell lifespan by 1.7 times, from 242 to 410 hours. This breakthrough paves the way for more durable, efficient fuel cell systems, potentially revolutionising technology in transport and power backup solutions.

The new skin sensor can help in rehabilitation of the sick and elderly

Researchers at Korea Advanced Institute of Science & Technology have developed a new electromyography (EMG) sensor that is stretchable and sticks to the skin, reliably capturing muscle signals for wearable robots, even when faced with sweat and skin debris. This technology is crucial for devices that assist in rehabilitation for the elderly and those recovering from strokes or injuries, as it accurately interprets movement intentions. Unlike traditional sensors that deteriorate or fail under challenging skin conditions, this microneedle sensor penetrates lightly into the skin, ensuring consistent, high-quality signal collection without skin prep. Its design allows it to flex with skin movement, minimising motion-related noise. In practical applications, it has successfully enabled a wearable robot to help users lift objects by precisely detecting muscle activity and converting it into action. It is promising to improve robot-assisted rehabilitation's effectiveness and precision.